HMS Community — Product Backlog

Project: Hospital Management System — Community Edition

Course: BM 321 Software Engineering

Responsible Role: Product Owner and Requirements Engineer

Status: Sprint 1 — Revised Product Backlog

1. Product Goal

The goal of HMS Community is to develop a web-based hospital management system that integrates patient registration, appointments, doctor and staff management, outpatient services, electronic medical records, laboratory operations, inpatient management, and user authorization with full audit logging.

The academic MVP (end of Week 8) will demonstrate the core patient workflow using synthetic data. The final release (Week 10) will extend this workflow with laboratory operations and administration features.

2. Epics (Main Modules)

ID	Epic	Scope
EP-01	Patient Management	Patient registration, patient profile, identity, contact and relative information.
EP-02	Appointment Management	Creating, rescheduling and cancelling appointments; doctor schedules.
EP-03	Doctor & Staff Management	Doctors, departments, duties and working hours.
EP-04	Outpatient Management	Outpatient admission, check-in and examination process.
EP-05	Electronic Medical Records (EMR)	Examination notes, diagnoses, medical history, allergies, medications and reports.
EP-06	Laboratory Management (LIS)	Lab test requests and laboratory results.
EP-07	Inpatient & Bed Management	Patient admission, discharge and bed occupancy tracking.
EP-08	Authentication, Authorization & Audit	Login, role-based authorization, account management and logging of all actions.

3. User Roles

Role	Main Responsibilities in the System
Administrator	Manages staff accounts, roles, permissions and departments; reviews the audit log.
Receptionist	Registers, searches and updates patients; creates, cancels and reschedules appointments; checks patients in.
Physician	Views daily schedule; conducts examinations; records notes, diagnoses and medical history; requests and reviews lab tests.
Nurse	Supports outpatient and inpatient care. Permissions to be defined in the role–permission matrix.

Role	Main Responsibilities in the System
Patient Care Assistant	Supports daily patient care. Permissions to be defined in the role–permission matrix.
Lab Technician	Views pending lab requests and enters results.
Cleaning Staff	Hospital cleaning personnel. Permissions to be defined in the role–permission matrix.

4. Prioritized Backlog Summary

Priorities follow the MoSCoW method: Must (required for the sprint goal), Should (important but can move to a later sprint), Could (if capacity allows), Won't (in scope of the product but not planned this term).

ID	Epic	User Story	Priority	SP	Sprint
US-01	EP-08	Staff login	Must	5	Sprint 2
US-02	EP-08	Role-based access control	Must	8	Sprint 2
US-03	EP-08	Logging of all actions	Must	5	Sprint 2
US-04	EP-08	View audit log	Should	3	Final
US-05	EP-08	Manage staff accounts and permissions	Should	8	Final
US-06	EP-01	Register patient	Must	8	Sprint 2
US-07	EP-01	Search patient	Must	3	Sprint 2
US-08	EP-01	Update patient information	Should	3	Sprint 2
US-09	EP-03	List doctors by department	Must	3	Sprint 2
US-10	EP-03	Define doctor working hours	Could	5	Final
US-11	EP-02	Create appointment	Must	8	Sprint 3
US-12	EP-02	Cancel appointment	Must	2	Sprint 3
US-13	EP-02	Reschedule appointment	Should	3	Sprint 3
US-14	EP-02	Physician daily schedule	Must	3	Sprint 3
US-15	EP-04	Start and complete examination	Must	3	Sprint 3
US-16	EP-04	Patient check-in	Could	2	Final
US-17.1	EP-05	Examination note and diagnosis	Must	5	Sprint 3
US-17.2	EP-05	Medical history and allergies	Must	3	Sprint 3
US-17.3	EP-05	Medication history and reports	Should	5	Sprint 3
US-18	EP-05	View patient history	Should	3	Sprint 3
US-19	EP-06	Request laboratory tests	Must (Final)	5	Final
US-20	EP-06	Enter laboratory results	Must (Final)	5	Final
US-21	EP-06	View laboratory results	Must (Final)	2	Final
US-22	EP-07	View bed occupancy	Could	5	Final
US-23	EP-07	Admit and discharge patient	Won't (this term)	8	Out of scope

5. User Stories and Acceptance Criteria

Each story lists its priority, story points (SP) and testable acceptance criteria. US-17 is split into three sub-stories so that each fits within one sprint.

EP-08 Authentication, Authorization & Audit

US-01 Staff login · Must · 5 SP
As a staff member, I want to log in with my username and password so that I can access the system according to my role.

- Valid credentials return a session/token and redirect the user to their role's home screen.
- Invalid credentials show an error message and no token is issued.
- The user can log out, and the token becomes invalid.

US-02 Role-based access control · Must · 8 SP
As an administrator, I want each function to be restricted by role so that users can only perform authorized actions.

- The system supports seven roles: Administrator, Receptionist, Physician, Nurse, Patient Care Assistant, Lab Technician, Cleaning Staff.
- The functions each role can access are defined in a role–permission matrix.
- An API request to a function outside the user's role returns HTTP 403.
- Menu items for unauthorized functions are hidden in the UI.

US-03 Logging of all actions · Must · 5 SP
As an administrator, I want every action performed in the system to be logged automatically so that all activity can be fully traced.

- All create, update, delete and cancel operations are logged.
- Viewing patient records and medical records is also logged.
- Successful and failed login attempts and logouts are logged.
- Each log entry contains user, role, action type, affected record ID and timestamp.
- For update operations, values before and after the change are stored.
- Log entries cannot be edited or deleted through the application.

US-04 View audit log · Should · 3 SP
As an administrator, I want to view and filter the audit log so that I can investigate system activity.

- Logs can be filtered by user, action type and date range.
- Only the Administrator can access this screen.

US-05 Manage staff accounts and permissions · Should · 8 SP
As an administrator, I want to manage staff accounts and role permissions within the system so that access can be fully controlled from the application.

- The administrator can create staff accounts, edit their details and deactivate them.
- The administrator can change a user's role.
- The administrator can reset a user's password.

- The administrator can edit the role–permission matrix to set which role can access which function.
- The administrator can add departments and assign doctors to departments.
- A deactivated user cannot log in.
- The administrator cannot deactivate their own account, and the last administrator account in the system cannot be removed.

EP-01 Patient Management

US-06 Register patient · Must · 8 SP
As a receptionist, I want to register a new patient so that their identity, contact and relative information are stored.

- Required fields: first name, last name, national ID (11 digits, synthetic), date of birth, gender, phone, address.
- A condition level is selected for the patient: Mild, Moderate or Advanced.
- For patients with an “Advanced” condition level, relative information is required: the relative’s full name, relationship degree and phone number.
- For patients with a “Mild” or “Moderate” condition level, relative information is optional.
- If a patient’s level is later raised to “Advanced” and no relative is registered, the system does not allow saving until relative information is entered.
- Missing or invalid required fields are rejected with a field-level error.
- A second patient with the same national ID cannot be registered.

US-07 Search patient · Must · 3 SP
As a receptionist, I want to search patients by national ID or name so that I can quickly find an existing record.

- Search by exact national ID returns the matching patient.
- Search by partial name returns all matching patients.
- An empty result shows a “no patient found” message.

US-08 Update patient information · Should · 3 SP
As a receptionist, I want to update a patient’s information so that their records stay current.

- All fields except the national ID can be updated.
- The validation rules of US-06 apply to updates as well.
- Each update is written to the audit log together with its before and after values.

EP-03 Doctor & Staff Management

US-09 List doctors by department · Must · 3 SP
As a receptionist, I want to view doctors filtered by department so that I can choose the right doctor for a patient.

- All departments are listed (seeded data).
- Selecting a department lists only its doctors.

US-10 Define doctor working hours · Could · 5 SP
As an administrator, I want to define each doctor's working hours so that appointment slots reflect real availability.

- Working days and hours can be set per doctor.
- The MVP uses seeded working hours until this story is completed.

EP-02 Appointment Management

US-11 Create appointment · Must · 8 SP
As a receptionist, I want to book an appointment in a doctor's available time slot so that the patient can be examined.

- Only 15-minute slots within the doctor's working hours are offered.
- A slot already booked for the same doctor cannot be booked again.
- Appointments cannot be created in the past or for an unregistered patient.

US-12 Cancel appointment · Must · 2 SP
As a receptionist, I want to cancel an appointment so that the slot becomes available to other patients.

- A cancelled appointment's slot becomes bookable again.
- Completed appointments cannot be cancelled.

US-13 Reschedule appointment · Should · 3 SP
As a receptionist, I want to move an appointment to another available slot so that patients can change their visit time.

- The same availability and overlap rules as US-11 apply.

US-14 Physician daily schedule · Must · 3 SP
As a physician, I want to see my appointments for today so that I know which patients to examine.

- The list shows only the logged-in physician's appointments, ordered by time.
- Each item shows patient name, time and status.

EP-04 Outpatient Management

US-15 Start and complete examination · Must · 3 SP
As a physician, I want to start and complete an examination from an appointment so that the visit status is tracked.

- Status flow: Scheduled → In Examination → Completed.
- Only the physician assigned to the appointment can start it.
- A cancelled appointment cannot be started.

US-16 Patient check-in · Could · 2 SP
As a receptionist, I want to mark a patient as arrived so that the physician knows who is waiting.

- Status changes to “Waiting” and appears on the physician’s list.

EP-05 Electronic Medical Records

US-17 Examination note, diagnosis and medical history · Must · 13 SP (total of sub-stories)
As a physician, I want to record and view a patient’s examination note, diagnosis and medical history during an examination so that I can evaluate the patient with their full health background.

This story is split into three sub-stories to fit within a single sprint:

US-17.1 Examination note and diagnosis · Must · 5 SP
As a physician, I want to record the complaint, findings and diagnosis of an examination so that the visit is documented.

- Complaint, findings and diagnosis are recorded; the note is linked to the patient and the examination.
- The diagnosis can be selected from a predefined ICD-10 subset or entered as text.
- The note can be edited until the examination is completed, then becomes read-only.

US-17.2 Medical history and allergies · Must · 3 SP
As a physician, I want to record and view a patient’s medical history and allergies so that I can avoid risks during treatment.

- Chronic conditions and past illnesses/surgeries are recorded.
- Allergies (drug, food, other) are recorded.
- Recorded allergies are shown as a prominent warning on the examination screen.

US-17.3 Medication history and reports · Should · 5 SP
As a physician, I want to record the patient’s current and past medications and their medical reports so that their treatment history is complete.

- Current and past medications are recorded with start and end dates.
- The physician can create medical reports (sick leave report, health report, etc.) or attach them as PDF.
- All of the patient’s reports are listed in chronological order.

US-18 View patient history · Should · 3 SP
As a physician, I want to view a patient’s previous examinations so that I can make informed decisions.

- Past examinations are listed in reverse chronological order with date, doctor and diagnosis.
- Only the Physician role can view clinical notes.

EP-06 Laboratory Management

US-19 Request laboratory tests · Must (Final) · 5 SP
As a physician, I want to request lab tests during an examination so that needed analyses are performed.

- Tests are selected from a predefined test catalog.
- A request can only be created for an examination that is in progress.
- New requests appear with status “Requested”.

US-20 Enter laboratory results · Must (Final) · 5 SP
As a lab technician, I want to see pending requests and enter results so that physicians receive them.

- Pending requests are listed with patient and test name.
- Entering a result changes status to “Completed”.
- Values outside the reference range are flagged.

US-21 View laboratory results · Must (Final) · 2 SP
As a physician, I want to view my patients’ lab results so that I can evaluate them.

- Results are shown under the related examination with reference ranges and flags.

EP-07 Inpatient & Bed Management

US-22 View bed occupancy · Could · 5 SP
As hospital staff, I want to see available and occupied beds by ward so that capacity can be monitored.

- Beds are listed by ward with their status (available or occupied).
- The number of available and occupied beds is shown for each ward.
- Only roles authorized in the role–permission matrix can access this screen.

US-23 Admit and discharge patient · Won’t (this term) · 8 SP
As a physician, I want to admit a patient to a bed and later discharge them so that inpatient stays are tracked.

- A patient can be admitted only to an available bed, and the bed becomes occupied.
- Discharging a patient records the discharge date and makes the bed available again.
- Admission and discharge are written to the audit log.

6. Dependencies and Technical Areas

Story	Dependencies	Technical Area	Responsible Roles	Assignet Owner
US-01	User and role data model; seeded staff accounts	Backend + Frontend + Data	Lead Software Architect and Technical Lead; Implementation and Integration Engineer; Product Owner and Requirements Engineer	HabibYazar huseyin-tutak ardaservetusta-lang

Story	Dependencies	Technical Area	Responsible Roles	Assignet Owner
US-02	US-01; role–permission matrix	Backend + Frontend	Lead Software Architect and Technical Lead; Implementation and Integration Engineer	HabibYazar ardaservetusta-lang
US-03	US-01	Backend + Data	Lead Software Architect and Technical Lead; Implementation and Integration Engineer; Product Owner and Requirements Engineer	HabibYazar huseyin-tutak ardaservetusta-lang
US-04	US-03	Backend + Frontend	Lead Software Architect and Technical Lead; Implementation and Integration Engineer	HabibYazar ardaservetusta-lang
US-05	US-02	Backend + Frontend + Data	Lead Software Architect and Technical Lead; Implementation and Integration Engineer; Product Owner and Requirements Engineer	HabibYazar huseyin-tutak ardaservetusta-lang
US-06	US-02, US-03	Backend + Frontend + Data	Lead Software Architect and Technical Lead; Implementation and Integration Engineer; Product Owner and Requirements Engineer	HabibYazar huseyin-tutak ardaservetusta-lang
US-07	US-06	Backend + Frontend	Lead Software Architect and Technical Lead; Implementation and Integration Engineer	HabibYazar ardaservetusta-lang
US-08	US-06, US-03	Backend + Frontend + Data	Lead Software Architect and Technical Lead; Implementation and Integration Engineer; Product Owner and Requirements Engineer	HabibYazar huseyin-tutak ardaservetusta-lang
US-09	Seeded department and doctor data	Backend + Frontend + Data	DevOps and Infrastructure Engineer; Implementation and Integration Engineer; Product Owner and Requirements Engineer	huseyin-tutak ardaservetusta-lang Zmly007
US-10	US-05, US-09	Backend + Frontend + Data	DevOps and Infrastructure Engineer; Implementation and Integration Engineer; Product Owner and Requirements Engineer	huseyin-tutak ardaservetusta-lang Zmly007
US-11	US-06, US-09; seeded working hours	Backend + Frontend + Data	Lead Software Architect and Technical Lead; Implementation and Integration Engineer; Product Owner and Requirements Engineer	HabibYazar huseyin-tutak ardaservetusta-lang
US-12	US-11	Backend + Frontend	Lead Software Architect and Technical Lead; Implementation and Integration Engineer	HabibYazar ardaservetusta-lang
US-13	US-11	Backend + Frontend	Lead Software Architect and Technical Lead; Implementation and Integration Engineer	HabibYazar ardaservetusta-lang

Story	Dependencies	Technical Area	Responsible Roles	Assignet Owner
US-14	US-11	Backend + Frontend + Data	Lead Software Architect and Technical Lead; Implementation and Integration Engineer; Product Owner and Requirements Engineer	HabibYazar huseyin-tutak ardaservetusta-lang
US-15	US-14	Backend + Frontend	DevOps and Infrastructure Engineer; Implementation and Integration Engineer	ardaservetusta-lang Zmly007
US-16	US-11	Backend + Frontend	DevOps and Infrastructure Engineer; Implementation and Integration Engineer	ardaservetusta-lang Zmly007
US-17.1	US-15; ICD-10 subset (seed)	Backend + Frontend + Data	Lead Software Architect and Technical Lead; Implementation and Integration Engineer; Product Owner and Requirements Engineer	HabibYazar huseyin-tutak ardaservetusta-lang
US-17.2	US-06, US-15	Backend + Frontend + Data	Lead Software Architect and Technical Lead; Implementation and Integration Engineer; Product Owner and Requirements Engineer	HabibYazar huseyin-tutak ardaservetusta-lang
US-17.3	US-17.2	Backend + Frontend + Data	Lead Software Architect and Technical Lead; Implementation and Integration Engineer; Product Owner and Requirements Engineer	HabibYazar huseyin-tutak ardaservetusta-lang
US-18	US-17.1	Backend + Frontend + Data	DevOps and Infrastructure Engineer; Implementation and Integration Engineer; Product Owner and Requirements Engineer	huseyin-tutak ardaservetusta-lang Zmly007
US-19	US-15; lab test catalog (seed)	Backend + Frontend + Data	DevOps and Infrastructure Engineer; Implementation and Integration Engineer; Product Owner and Requirements Engineer	huseyin-tutak ardaservetusta-lang Zmly007
US-20	US-19	Backend + Frontend	DevOps and Infrastructure Engineer; Implementation and Integration Engineer	ardaservetusta-lang Zmly007
US-21	US-20	Backend + Frontend	DevOps and Infrastructure Engineer; Implementation and Integration Engineer	ardaservetusta-lang Zmly007
US-22	Seeded ward and bed data	Backend + Frontend + Data	Lead Software Architect and Technical Lead; Implementation and Integration Engineer; Product Owner and Requirements Engineer	HabibYazar huseyin-tutak ardaservetusta-lang
US-23	US-22	Deferred	Lead Software Architect and Technical Lead; Implementation and Integration Engineer	HabibYazar ardaservetusta-lang

Actual GitHub assignees will be confirmed by the team during sprint planning. The Product Owner maintains backlog prioritization and acceptance criteria; technical owners carry out implementation, the QA engineer verifies acceptance criteria through automated tests, and the DevOps engineer maintains the CI/CD pipeline and Docker deployment.

7. Release Plan and Sprint Scope

Sprint	Must Stories	Should / Could Stories	Must SP
Sprint 1 (Weeks 1–2)	No product stories. Enablers: GitHub organization, Git-Flow rules, docker-compose skeleton, draft OpenAPI contract, initial backlog.	—	—
Sprint 2 (Weeks 3–5)	US-01, US-02, US-03, US-06, US-07, US-09	US-08	32
Sprint 3 (Weeks 6–8) — MVP	US-11, US-12, US-14, US-15, US-17.1, US-17.2	US-13, US-17.3, US-18	24
Final (Weeks 9–10)	US-19, US-20, US-21	US-04, US-05; Could: US-10, US-16, US-22	12
Out of scope	—	US-23	—

MVP workflow (Sprint 3): Login → Patient Registration → Appointment → Physician Daily Schedule → Examination → Examination Note, Diagnosis and Medical History.

Final workflow (Week 10): MVP workflow → Laboratory Request → Laboratory Result → Physician Reviews Result → Administrator Reviews Audit Log.

Laboratory stories (US-19 to US-21) are planned for the final release so that the Week 8 MVP can meet the 50 percent unit-test coverage requirement with a stable core workflow. If team velocity is lower than planned, Should and Could stories are moved to the next sprint first; Must stories are not removed.

8. Definition of Ready

A user story is ready when:

- The user and expected functionality are clearly defined.
- Acceptance criteria are testable.
- Dependencies are identified.
- Story points are estimated.
- A responsible team member is assigned.
- The story is small enough for one sprint (maximum 8 SP).

9. Definition of Done

A user story is completed when:

- The required functionality is implemented.
- All acceptance criteria are satisfied and accepted by the Product Owner.

- Relevant automated tests pass and unit-test coverage does not fall below 50 percent.
- The code is reviewed and merged through a Pull Request.
- The OpenAPI contract and documentation are updated when necessary.
- The completed work is traceable through GitHub Issues, Pull Requests and commits.

10. Notes, Assumptions and Open Issues

- Story points are initial estimates and may be revised during sprint planning.
- Only synthetic hospital and patient data will be used for development and demonstration.
- The eight epics represent the Community system scope; not all modules are expected to be fully implemented this term. Patient referral (EP-04), sample tracking (EP-06) and ward management (EP-07) are not covered by the current stories.
- Assumption: the patient condition level in US-06 is set by the Receptionist. If this decision belongs to the Physician, the field and its permission will move to the examination screen.
- Open issue: the Nurse, Patient Care Assistant and Cleaning Staff roles do not yet have dedicated user stories. Their functions must be defined before Sprint 2 planning so that each role has a usable screen in the demonstration.